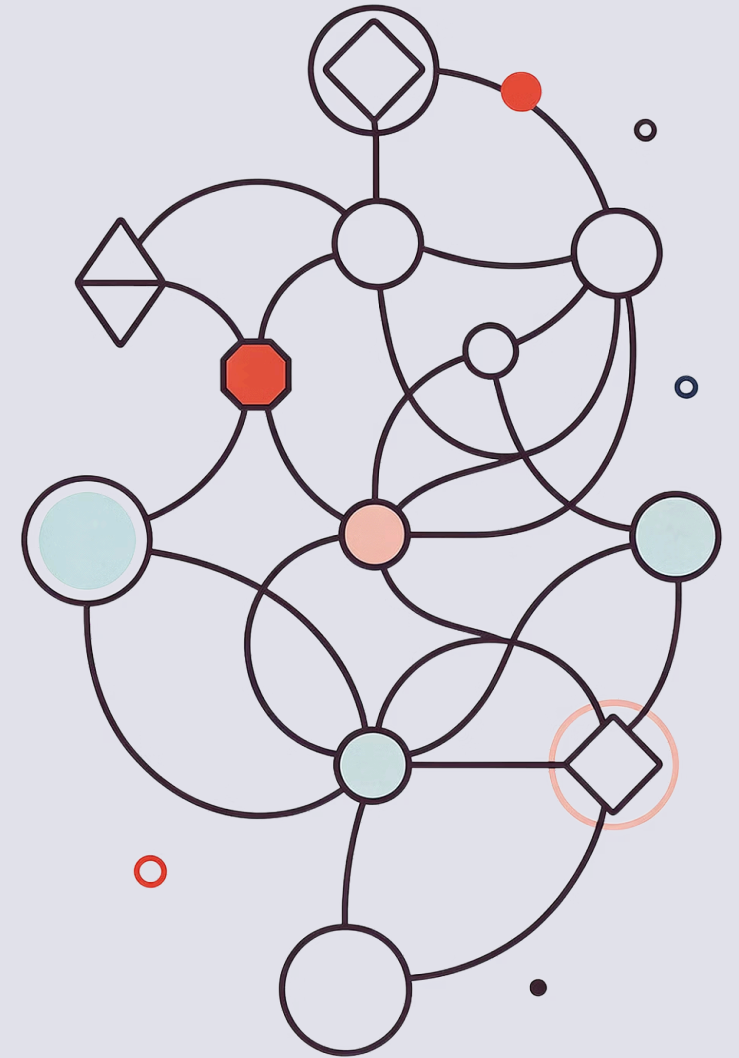
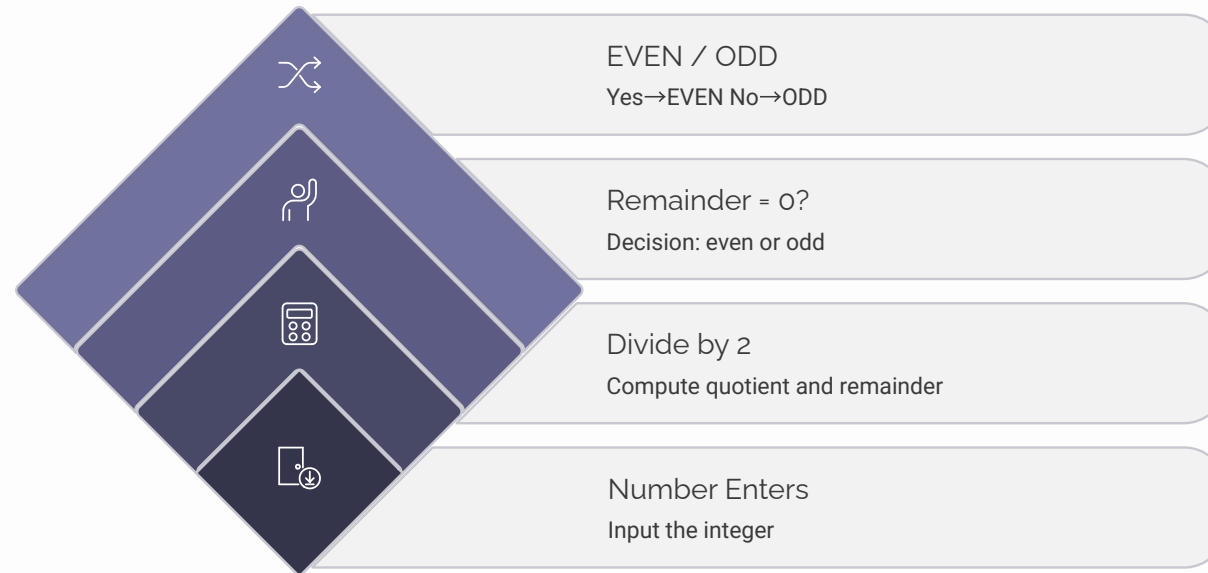


From Rules to Learning: Why Machine Learning Needs Mathematics

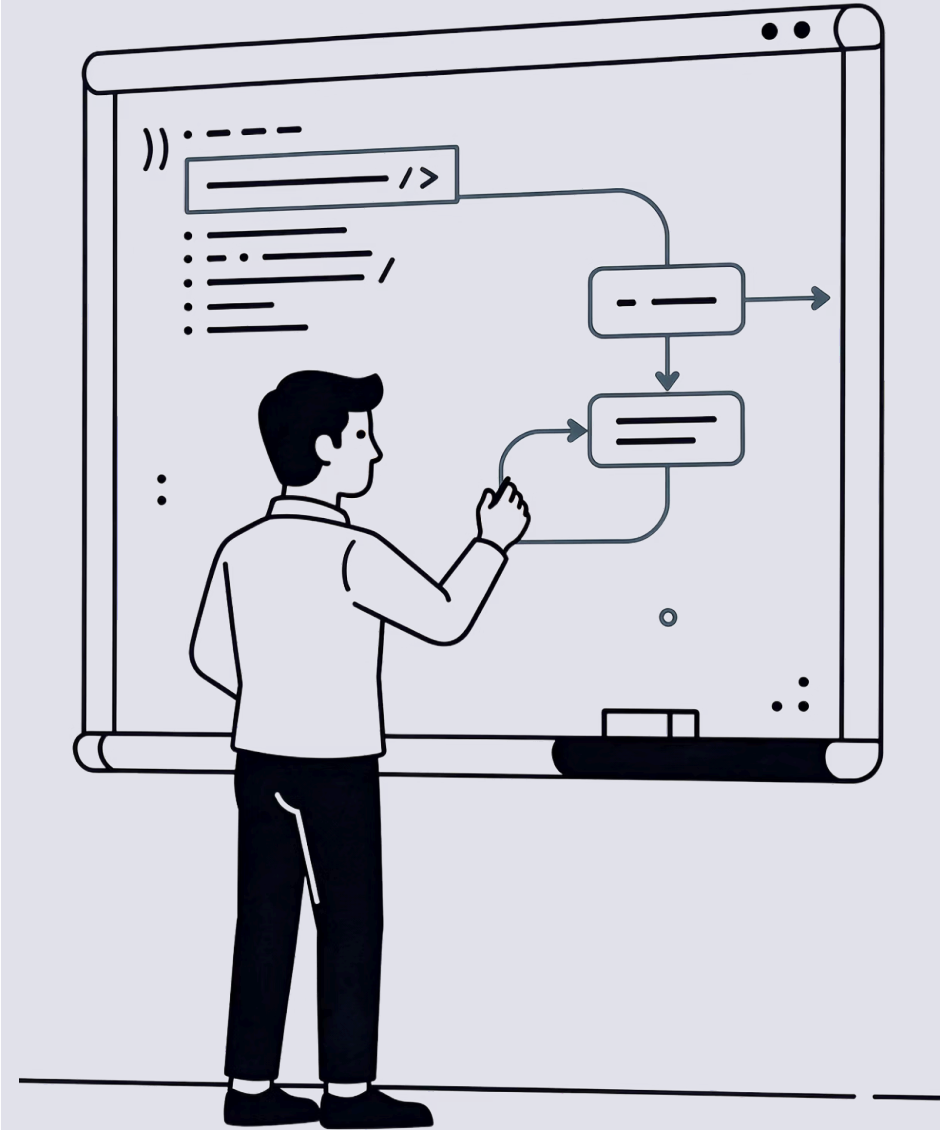
A visual introduction for engineering students



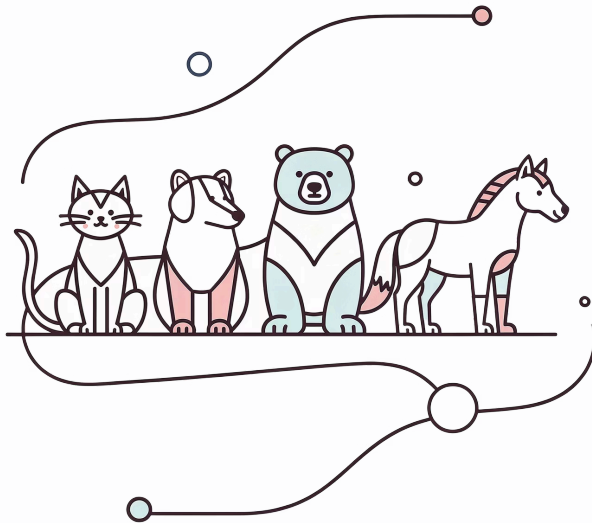
The Old Way: Rule-Based Programming



The programmer writes every rule by hand — logic must be explicit and complete before the program can run.



Where Rules Break Down: The Cat Problem



Rules that seem to work... don't

Four legs

Also: dog, horse, bear, fox...

Fur

Also: dog, bear, rabbit...

Pointed ears

Also: fox, dog...

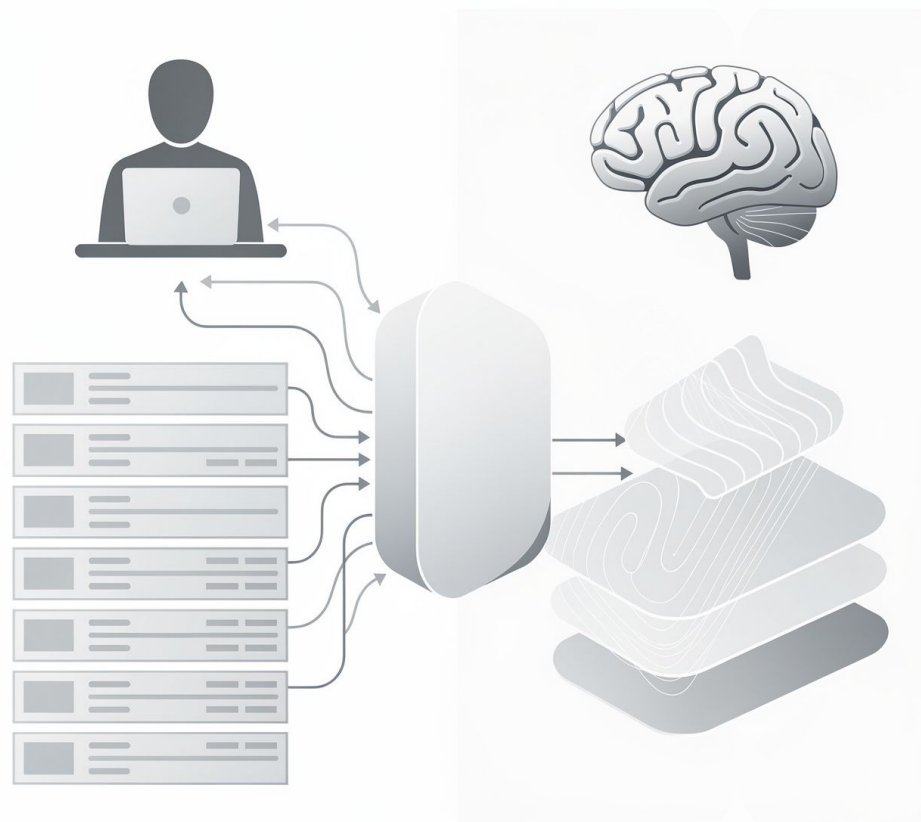
Breeds, poses, lighting

Rules explode in complexity

The Shift to Machine Learning

Old Way

Data + Rules →
Output
(programmer)

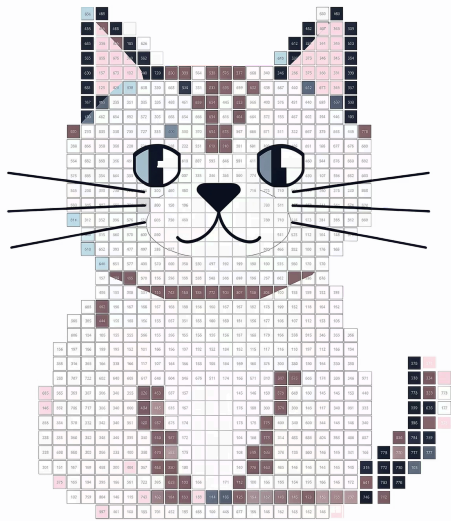


ML Way

Data + Examples →
Model → Prediction
(brain)

Show the model thousands of labeled examples — it discovers patterns automatically. No manual rules needed.

But What Does "Learning" Actually Mean?



Computers only understand numbers



Image

A photograph of a cat



Pixel Grid

Every pixel has a value 0–255



Number Matrix

ML models operate on these values



Learning

Adjusting numbers inside the model

Where Does the "Matrix" Come In?



The model learns **weights** — numerical parameters that transform inputs into predictions. Vectors, matrices, and tensors are the containers.

Why Mathematics Enters Machine Learning

"If the machine learns from numbers — how does it measure patterns, mistakes, and improvements?"



Linear Algebra

Represent and transform data



Probability & Statistics

Uncertainty, distributions, patterns



Calculus

How changing parameters affects error



Optimization

Minimize error, improve the model

The Gradient Descent Analogy



Model = Hiker on a Mountain

Loss = Altitude

Height represents how wrong the model is

Gradient = Slope Direction

Steepest direction of climb

Negative Gradient = Downhill

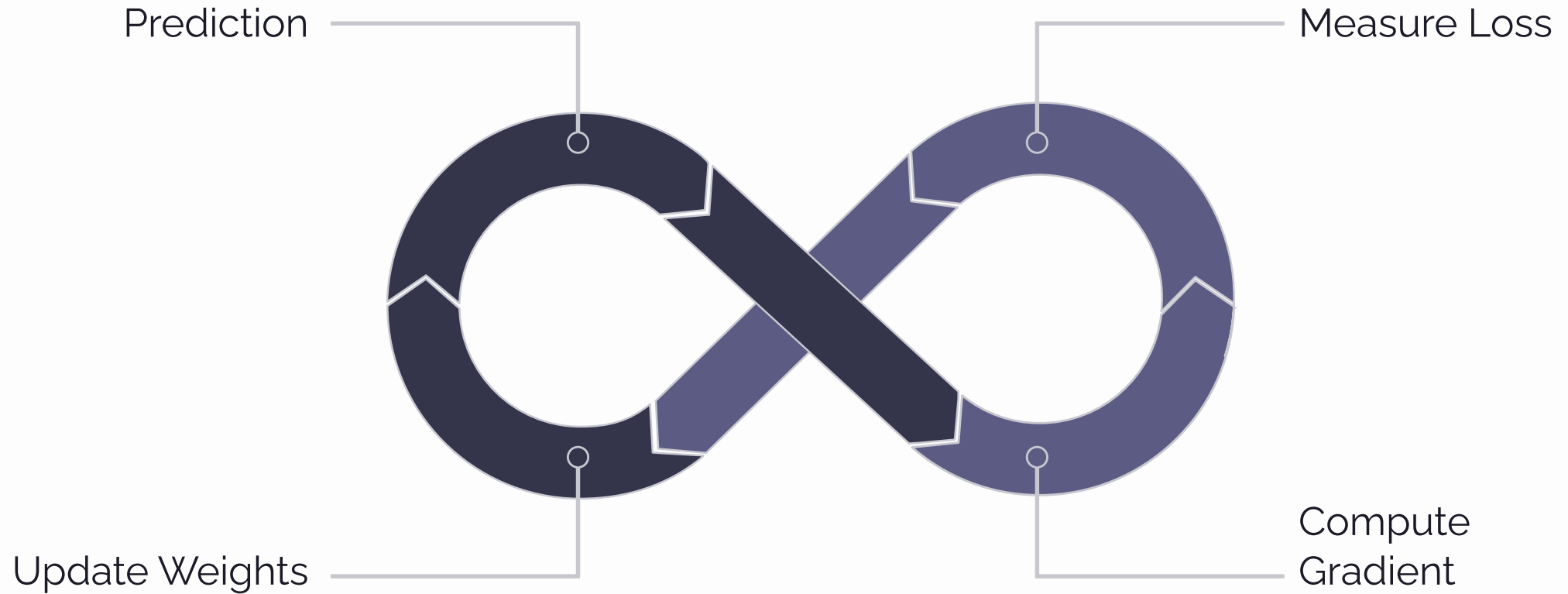
Direction toward less error

Learning Rate η = Step Size

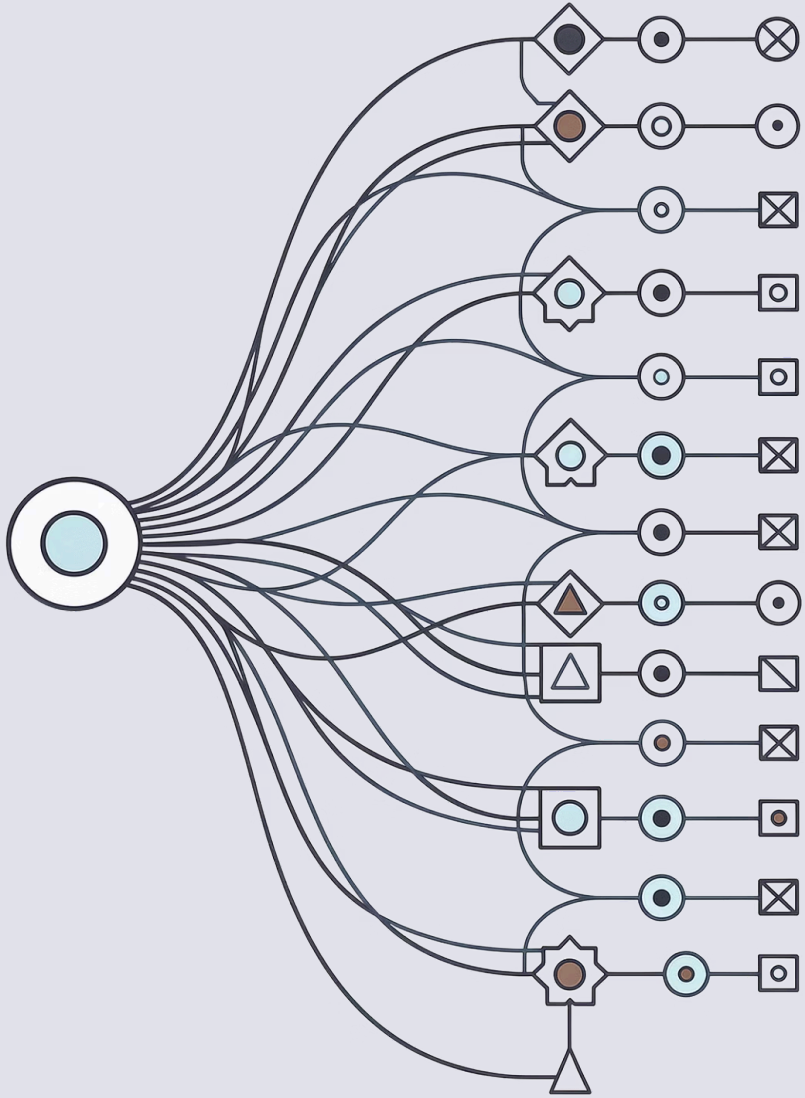
How large each step is

$$w_{\text{new}} = w_{\text{old}} - \eta \frac{\partial L}{\partial w}$$

The Learning Loop



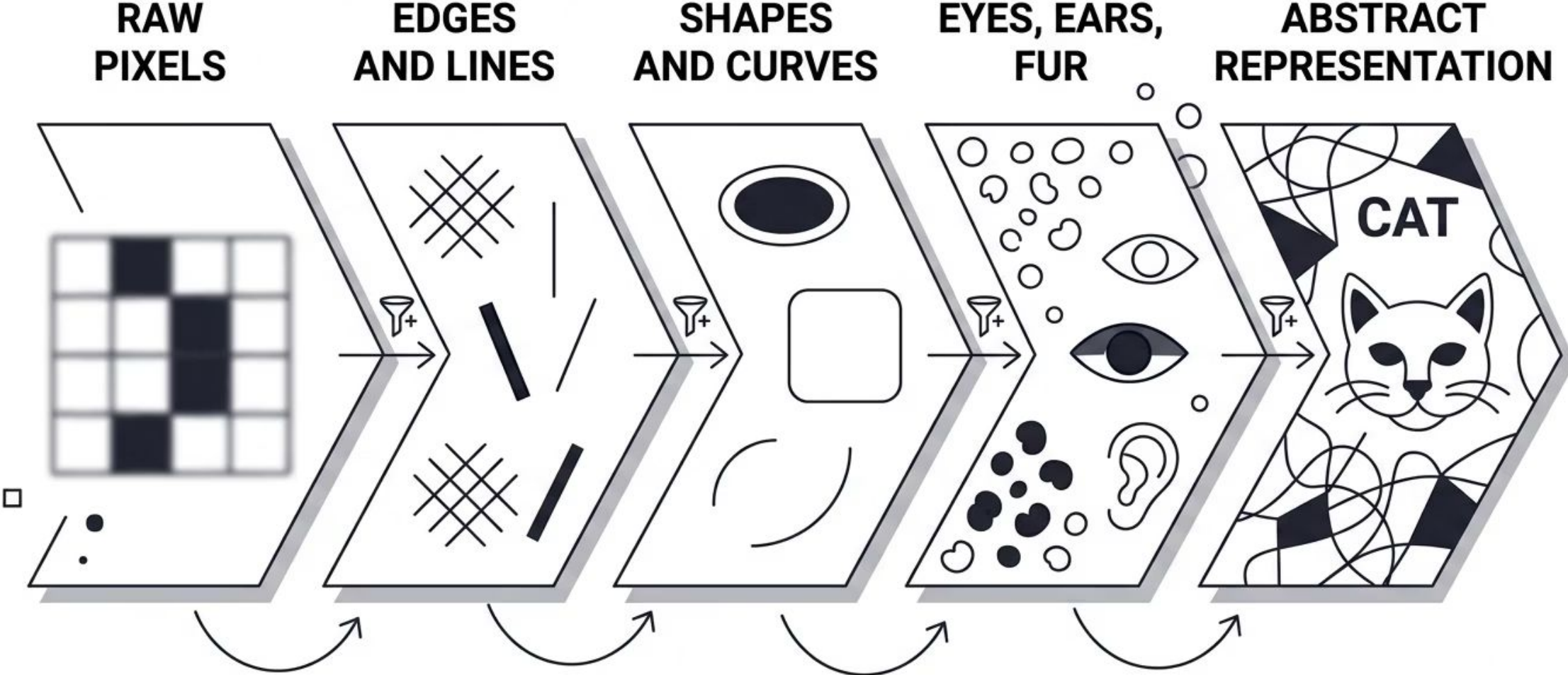
Each cycle is one step downhill toward a smarter model. This loop is the engine of **all** machine learning.



From One Neuron to Deep Learning

- 1 Single Neuron
Inputs $\rightarrow$ Weighted Sum + Bias $\rightarrow$ Activation $\rightarrow$ Output
- 2 A Layer
Stack neurons side by side
- 3 Neural Network
Stack layers: input $\rightarrow$ hidden $\rightarrow$ output
- 4 Deep Network
Many hidden layers = Deep Learning

How a CNN Learns to Recognize a Cat



Mathematics Behind Common ML Algorithms

Linear Regression

Linear Algebra + Statistics

Logistic Regression

Probability + Calculus

K-Means Clustering

Distance + Optimization

PCA

Linear Algebra
(Eigenvectors)

Decision Trees

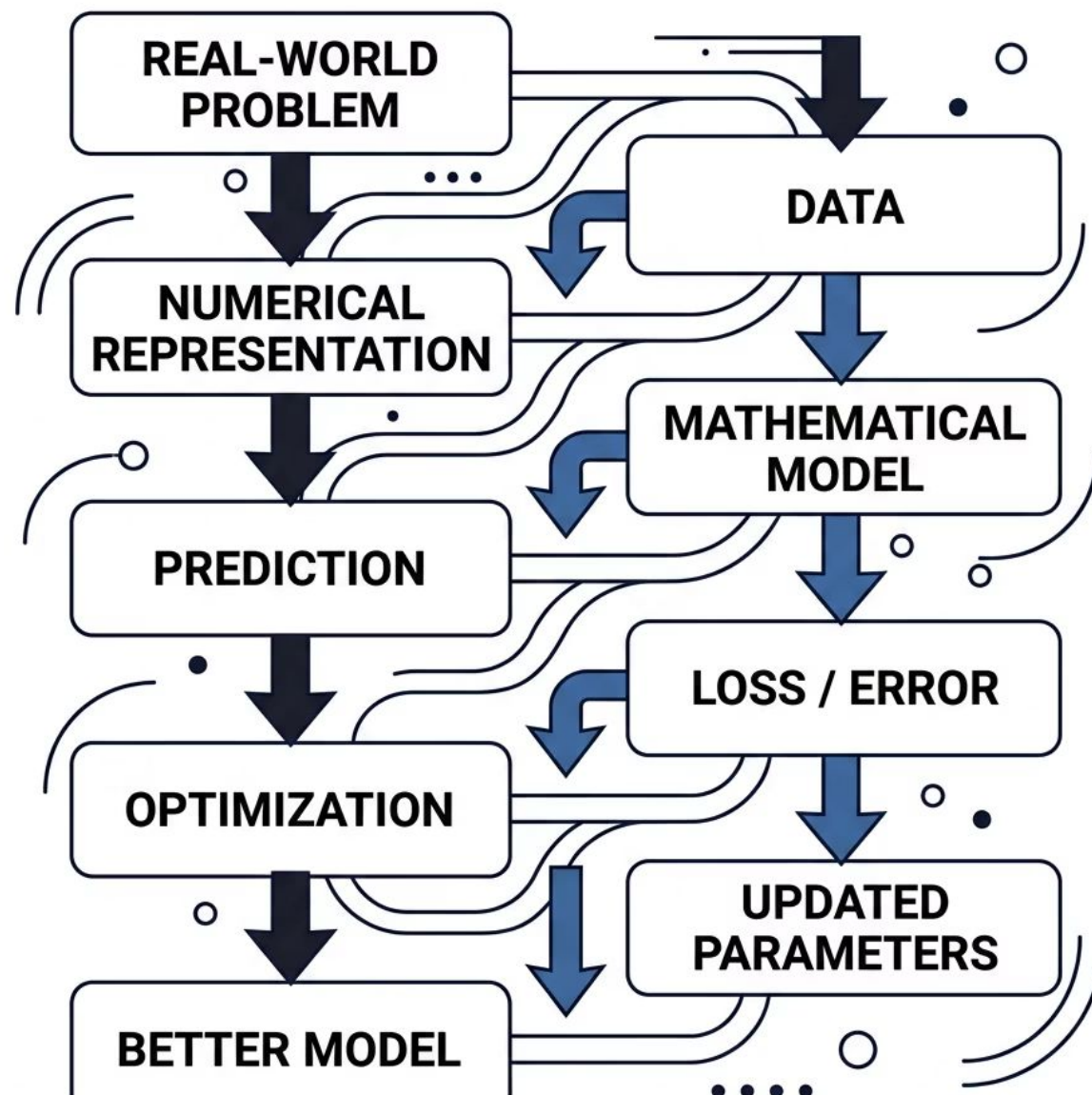
Entropy + Information Theory

Neural Networks

Linear Algebra + Calculus + Optimization

The diagram shows a directed graph with the following components and connections:

- Nodes:**
 - A rectangular node on the left with an incoming arrow from the left.
 - A circular node in the center.
 - A tall cylindrical node at the top right.
 - A medium cylindrical node at the top center.
 - A medium cylindrical node at the top left.
 - A tall cylindrical node in the middle.
 - A medium cylindrical node on the right.
 - A medium cylindrical node at the bottom left.
 - A medium cylindrical node at the bottom right.
- Connections:**
 - The left rectangular node connects to the center circular node and the top-left cylindrical node.
 - The center circular node connects to the top-left cylindrical node, the top-center cylindrical node, the middle cylindrical node, and the bottom-left cylindrical node.
 - The top-left cylindrical node connects to the top-center cylindrical node.
 - The top-center cylindrical node connects to the top-right cylindrical node and the middle cylindrical node.
 - The top-right cylindrical node connects to the middle cylindrical node.
 - The middle cylindrical node connects to the right cylindrical node and the bottom-right cylindrical node.
 - The right cylindrical node connects to the bottom-right cylindrical node.
 - The bottom-left cylindrical node connects to the middle cylindrical node and the bottom-right cylindrical node.
 - The bottom-right cylindrical node has an outgoing arrow to the right.



The Final Takeaway



Rule-Based Programming

Tells the computer what rules to follow



Machine Learning

Allows the computer to learn patterns from examples



Mathematics

The language to represent data, measure error, and optimize learning

"Don't memorize the mathematics. Understand why the mathematics is needed."